BSAID

<Version 1>

by

Martin, Aliyah Vhallery R.

Bernesto, Vince Patrick S.

Camayang, Geraldine M.

Castro, Criselle Mae T.

Guintu, Jona Mae M.

Landingin, Neil Joseph

Mercolesia, Mikaella Nicole P.

Sam, Cedric Isaac C.

<<June 4, 2025>>

TABLE OF CONTENTS

TITLE	PAGE
Title Page	i
Approval Sheet	ii
Table of Contents	iii
List of Tables	iv
List of Figures	v

Chapter

1 PROJECT OVERVIEW

- Logline
- Background of the Project
- Project Design Goals
- Scope and Limitations of the Project
- Game Overview
  - Core Gameplay
  - Genre
  - Target Audience
  - Target Platform
  - Review of Related Game/Literature

2 GAME STORY AND NARRATIVE

- Theme
- Synopsis
- Screenplay
- Character
- Settings
- Environment
- Plot
- Storyboard

3 GAMEPLAY

Level Design

Game Progression and Level Outline

Level Objectives

Level Mechanics

Physics

Movement in the Game

Objects

Actions

Combat

Economy

Screen flow

Game Options

Replaying and Saving

Easter Eggs and Cheats

User Interface

GUI Layout

Camera Set Up

Icons

Game Controls

Help Systems

Game Feature

Development Timeline

4. EVALUATION AND TESTING

Hardware and Software Requirements.....

Development

User

Game Screenshots .....

Summary of Evaluation Result.....

Functionality Testing

Beta Testing

Results

Appendices

Complete Story

Multimedia Bible

Operational Definition of Terms

User's Manual / Game Playbook

Game Installer and Installation Procedures…….

Final Game Poster……..

*Hardware and Software Requirements

*Development

*User

Chapter 1 – PROJECT OVERVIEW

Logline

Every transaction tells a story. Will you make the right entry?

Background of the Project

BSAid was born from the experiences of the proponents as Accountancy students who once stood exactly where today's freshmen are. Having endured countless hours of studying, moments of self-doubt, and the challenges of understanding accounting concepts for the first time, the proponents understand how overwhelming the beginning of the BSA journey can be. Over the years, they witnessed classmates and underclassmen struggle with the same difficulties, and some eventually chose to leave the program altogether. As they approached the end of their own journey, the proponents wanted to create something meaningful that could help future generations of Accountancy students overcome the challenges they once faced. With the help of technology professionals, they developed BSAid, a simulation-based learning platform that transforms accounting concepts into interactive and engaging experiences, with the hope of making learning more accessible, enjoyable, and less intimidating for aspiring accountants.

Project Design Goals

The development of BSAid is guided by the following goals:

• To create a simulation-based educational game that introduces accounting concepts through interactive gameplay.

• To transform fundamental accounting topics into engaging challenges, quests, and learning activities.

• To provide players with realistic accounting scenarios that encourage critical thinking and decision-making.

• To incorporate immediate feedback mechanisms that allow players to learn from their actions and improve their understanding of accounting concepts.

• To implement progression systems, rewards, and achievements that motivate players to continue learning and exploring the game.

• To design an accessible and user-friendly platform that can be utilized by first-year Accountancy students as a supplementary learning tool.

• To develop a game experience that balances educational content and entertainment to promote sustained player engagement.

• To establish a platform that can be expanded and improved through future content updates and user feedback.

Scope and Limitations of the Project

BSAid is a simulation-based educational game designed primarily for first-year Bachelor of Science in Accountancy students. The game focuses on selected topics under Fundamentals of Accounting and Financial Accounting and Reporting 1 (FAR1), including accounting principles, business transactions, journal entries, adjusting entries, and other introductory accounting concepts.

The platform is intended to serve as a supplementary learning tool that reinforces classroom instruction through interactive and game-based experiences. It is not designed to replace lectures, textbooks, classroom discussions, or other formal learning materials used in the course.

The current version of the game is limited to the content, mechanics, and features implemented during the development period. While the platform incorporates

realistic accounting scenarios and simulations, it does not encompass all areas of accounting education or professional accounting practice. Future versions may include additional topics, expanded gameplay mechanics, enhanced user features, and other improvements based on user feedback and development opportunities.

GAME OVERVIEW

Core Gameplay

BSAid is a simulation-based educational game that uses interactive gaming to teach students basic accounting concepts. Players assume the character of an aspiring accounting student who faces challenges with financial and business-related situations. The game uses a progressive learning approach that unlocks lessons and awards as players show knowledge of accounting topics. It features accounting transactions, quizzes, mini-games, and decision-based exercises that are in line with the FAR1 curriculum.

Genre

BSAid combines components from several game genres, such as:

- Educational Game (Edutainment)
- Simulation
- Visual Novel
- Puzzle
- Role-Playing Game (RPG)

The platform can present accounting ideas through storytelling, exploration, problem-solving, character interaction, and interactive simulations through the combination of several genres. This combination preserves educational value while producing a more immersive and captivating learning environment.

Target Audience

First-year Bachelor of Science in Accountancy (BSA) students enrolled in Fundamentals of Accounting and Financial Accounting and Reporting 1 (FAR1) are the main target audience for BSAid. By offering interactive opportunities to learn and reinforce fundamental accounting concepts outside of the classroom, the platform is intended to assist learners in the field of accounting.

Students studying accounting, business, and management (ABM), teachers looking for additional teaching resources, students getting ready for tests and quizzes, and anybody interested in learning the principles of accounting through interesting digital experiences may all utilize BSAid.

Target Platform

In order to guarantee accessibility, convenience, portability, and compatibility with devices frequently used by students, BSAid is primarily made for Android mobile devices.

Depending on user demand and development viability, future platform extension may potentially be taken into consideration.

Review of Related Games and Literature

Papers, Please

Papers, Please is a simulation game developed by Lucas Pope that places players in the role of an immigration inspector responsible for evaluating documents and making decisions based on established rules and procedures.

Minecraft: Education Edition

Minecraft: Education Edition is an educational adaptation of the sandbox game Minecraft developed by Mojang Studios. The platform allows students to engage in

collaborative problem-solving, exploration, and creativity within interactive virtual environments.

Kahoot!

Kahoot! is a game-based learning platform that utilizes quizzes, competitions, and interactive activities to enhance classroom participation and student engagement.

Grocery Cashier Games

Grocery Cashier Games are simulation-based games in which players take on the role of a cashier responsible for handling customer transactions, calculating totals, processing payments, and managing store-related tasks within a virtual environment.

Relevance to BSAid

Combined, the games and platforms under evaluation show how simulation-based learning, gamification, interactivity, and procedural gaming can enhance student motivation, engagement, and comprehension. Examples include Kahoot and Minecraft: Education Edition!Please demonstrate how immersive environments, tests, transactional tasks, and rule-based decision-making can transform difficult or repetitive learning content into captivating experiences that promote active engagement and skill improvement in grocery cashier simulation games and papers.

In a manner comparable, BSAid embraces these ideas by incorporating gamified advancement, interactive exercises, procedural problem-solving, instant feedback systems, and simulation dynamics to make accounting concepts more interesting and student-centered. This idea that educational content can be effectively given through immersive digital experiences that encourage ongoing learning, real-world application, and deeper understanding is supported by these related technologies.

Chapter 2 – GAME STORY AND NARRATIVE

Theme

A Visual-Novel Style game centered around Learning Accountancy and the Different Experiences and Stories of College Life

Game Narrative

The narrative focuses on the theme of different people having different stories in their lives. The narrative features one main storyline that also has three other branching storylines each with their own set of characters. After the first stage, the player can make a choice as to what branching storyline they will choose to be a part in that will play out after completing each stage.

Main Storyline: Your childhood friend and current schoolmate Faraday decided to enter the University's Accountancy Program because Faraday wanted to follow the steps of his sister Farahnaz. Things took a turn when Farahnaz found out that Faraday enrolled in the Accountancy Program

Branch 1: Your classmate Baxter is urging Macy to focus more on her studies so that she could still continue her studies under scholarship. Despite trying her best, Macy failed most of her quizzes in FAR2 and is left wondering what happened to her life that made her unable to reach the same high scores she used to obtain back in elementary and high school. (Talks about "Gifted Kid Burnout Syndrome", College Course Mismatch, and Mental Health Problems)

Branch 2:

Branch 3: The School's Assistant Librarian Philip Defensor offers to play one of the board games provided by the Library with you after Faraday had to rush off to do a favor for a friend. Whilst playing, the two of you talked about topics ranging from your decision to (Paranormal Activity, Lingering Regrets, and Moving On. "Bakit may multo?" para mas thrilling)

Screenplay

(To be discussed in Version 2)

Character

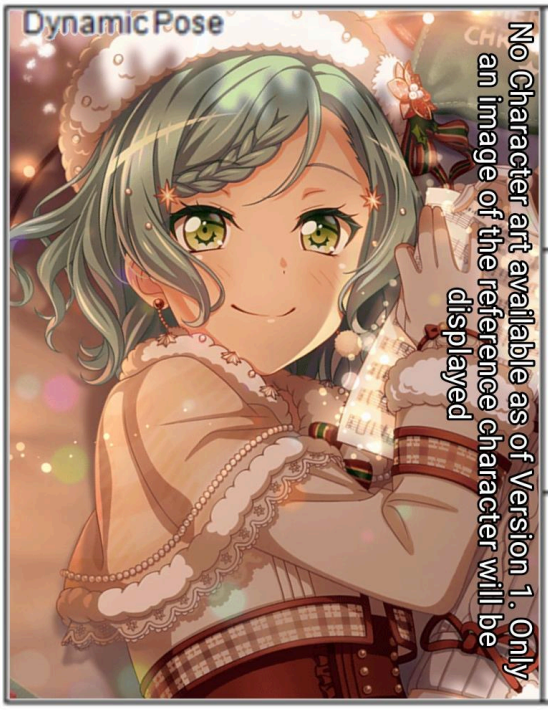
Characters Introduced in Main Storyline:

Character 1 : Main Character	
Dynamic Pose  Note: Future Version of the Game Design Document may have the option for the player to choose a design they want for the Main Character	Facial Expression Equipments/Accessories Other Pose

Background	Based on most visual-novel style games that have a generic character design meant to immerse the player through that character.
Personality	Blank, mostly reactive to situations. They are supposed to be a representative of the game's player that allows them to project themselves to the Main Character
History	A regular freshmen student who went into the accountancy course after their childhood friend begged them to join him

Character 1.1 : Faraday

Dynamic Pose



No Character art available as of Version 1. Only an image of the reference character will be displayed

Facial Expression

Equipments/Accessories

Other Pose

Background	Faraday will be heavily inspired by Hina Hikawa's personality of being a quirky natural genius and the Accounting Board Subject of FAR. Much like how FAR will become your first Accounting-related subject on your freshmen years, Faraday will be the first character you will meet and will be the first character involved with the first storyline of the game
Personality	Faraday is a very optimistic and adventurous individual. He has been regarded as a born genius for being able to master different kinds of skills in an incredibly short time, yet he never seems to be able to understand the sentiment of other people having trouble in things he finds easy. Because of this, he never really had many friends because of his quirky and candid behavior and his "geniusness" causing others to feel envy or jaded towards him. Regardless, he will always try his best in helping other people as long as they ask him for it directly.
History	A freshmen student who was regarded to be a born genius despite his quirkiness. He has an older sister named Farahnaz and is childhood friends with the Main Character.

Character 1.2 : Farahnaz

Dynamic Pose



No Character art available as of Version 1. Only an image of the reference character will be displayed

Facial Expression

Equipments/Accessories

Other Pose

Background	Farahnaz will be heavily inspired by Sayo Hikawa's personality of being a strict and disciplined hardworker and the Accounting Board Subject of AFAR. Much like how AFAR is connected to FAR by introducing advanced accounting procedures for different business settings in the professional world like governmental, corporation, construction, and even foreign currency translation, Farahnaz will be introduced as an alumni who have received several letters of recruitment from well-respected accounting firms. Faraday also shows great appreciation towards her to represent how connected the course subjects they represent
Personality	Compared to Faraday, Farahnaz is somewhat cold and stiff in nature. While Faraday who can master skills quickly, Farahnaz relentlessly hones her skills through rigorous practice, resulting in her being the type of person who never cuts corners. The difference between the two siblings has led to Farahnaz developing an inferiority complex, eventually growing distant to her younger brother.
History	A well-respected alumni who ranked first in the most recent CPA Licensure Examination. Despite graduating from college, she still interacts regularly with her underclassmen and her past professors. It was through them that she heard about her younger brother entering the Accountancy Program

Settings

The primary setting of BSAid is Pamantasan ng Lungsod ng Valenzuela (PLV), where the player assumes the role of a first-year Accountancy student. The game takes place across various locations commonly associated with student life and academic learning, creating a familiar and relatable environment for its intended audience.

Key locations may include:

- **University Campus**

  The central hub of the game where players navigate different areas, interact with characters, access learning activities, and progress through the storyline.

- **Classrooms**

  Learning environments where accounting lessons, discussions, and educational activities are introduced. Classrooms serve as the primary location for topic-based learning and instructional events.

- **Library**

  A study-oriented environment where players may access supplementary materials, review previously learned concepts, and participate in optional learning activities.

- **Student Areas**

  Common areas within the campus where players may interact with NPCs, receive quests, gather information, and experience aspects of student life beyond academics.

- **Simulation Areas**

  Specialized game spaces designed for accounting-related challenges, exercises, and mini-games that allow players to apply accounting concepts in practical scenarios.

Environment

BSAid features a school-based environment that combines elements of academic life with interactive educational gameplay. The overall atmosphere is designed to be welcoming, engaging, and supportive, encouraging players to explore, learn, and progress at their own pace.

The visual environment reflects a modern university setting inspired by Pamantasan ng Lungsod ng Valenzuela, incorporating recognizable academic spaces, student facilities, and learning environments. The game utilizes a clean and accessible design to ensure that educational content remains the primary focus while maintaining an enjoyable player experience.

Learning activities are integrated naturally into the environment through quests, classroom interactions, simulations, and mini-games. As players progress, the environment evolves to reflect their academic growth, providing a sense of advancement and accomplishment throughout their journey.

The environment aims to balance realism and engagement by recreating familiar aspects of student life while introducing interactive elements that support the game's educational objectives.

Environment _____ : _____

Concept Art	Structure
	Vegetation
	Aesthetics and other game objects
Background	
History	

(Images to follow on Version 2)

Plot

(To be discussed in Version 2)

Storyboard

(To be discussed in Version 2)

Chapter 3 – GAME PLAY

Game Progression and Level Outline

Outlines visually and discusses how the whole game is modeled from starting the game to its ending.

Case #01: Sari-Sari Store (Debit-Credit & Journalizing)

- **Level 1**
  - The player will answer multiple choice theory questions about financial transactions, debit & credit entries and Journalizing
- **Level 2**
  - The player will be given line items and they will answer its normal balance whether it is debit or credit
- **Level 3**
  - The player will be given transactions and they will have to analyze which account will be credited or debited
- **Level 4**
  - The player will be given transactions and they will have to create a journal entry for the given transaction

Special Case: Journal Jamboree

- **Sp#1**
  - The player will answer a variety of questions about Journalizing with medium difficulty
- **Sp#2**
  - The player will answer a variety of questions about Journalizing with hard difficulty

Case #02: Coffee Shop (Posting & Unadj. Trial Balance)

- **Level 1**
    - The player will answer multiple choice theory questions about posting and trial balances
- **Level 2**
    - The player will be given transactions and they will have to post a journal entry for the given transaction using a T account
- **Level 3**
    - The player will be given problems regarding the accounting equation (Assets - Liability = Equity)
- **Level 4**
    - The player will be given T accounts and will create an unadjusted trial balance based on the transactions given.

Special Case: Balancing the Books

- **Sp#3**
    - The player will be given a short comprehensive problem and will create an unadjusted trial balance based on the problems given.
- **Sp#4**
    - The player will be given a long comprehensive problem and will create an unadjusted trial balance based on the problems given.

Case #03: Jewelry Store (Adjusting Entries)

- **Level 1**
    - The player will answer multiple choice theory questions about adjusting entries
- **Level 2**
    - The player will be given Accrual transactions and they will have to analyze which line items to adjust or leave be
- **Level 3**
    - The player will be given Deferral transactions and they will have to analyze which line items to adjust or leave be.
- **Level 4**
    - The player will be given a trial balance and will have to adjust certain accounts based on the problems.

Special Case: Apparent Adjustments

- **Sp#5**
    - The player will be given short comprehensive problems and will have to analyze the given transactions and what adjustments need to be done
- **Sp#6**
    - The player will be given short comprehensive problems and will have to analyze the given transactions and what adjustments need to be done

Case #04: Supermarket (Adjusted Trial Balance & Worksheet)

- **Level 1**

    - The player will answer multiple choice theory questions about an adjusted trial balance and worksheet
- **Level 2**
    - The player will be given adjustments and they will have to reflect it onto the unadjusted trial balance.
- **Level 3**
    - The player will be given adjusted trial balances and they will have to create a worksheet in order to close the books.

Special Case: Worksheet Worries

- **Sp #7**
    - The player will be given short comprehensive problems and they will have to analyze adjustments in order to create the adjust trial balance and transfer it into a worksheet.
- **Sp #8**
    - The player will be given long comprehensive problems and they will have to analyze adjustments in order to create the adjust trial balance and transfer it into a worksheet.

Level Objectives

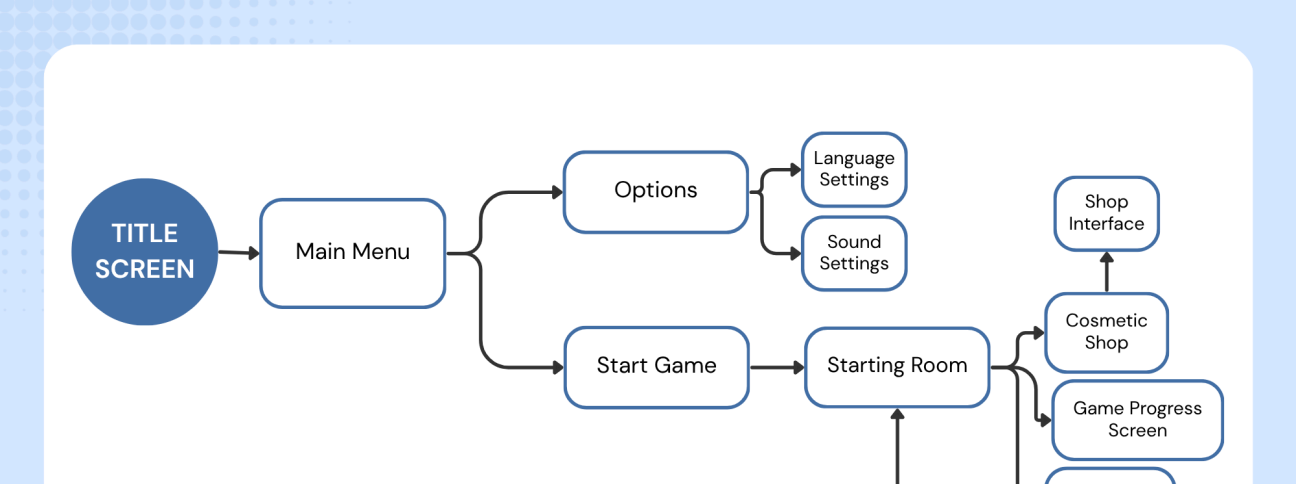
The player will be given a set of case files of varying difficulty. Each case file can be accessed right away but it is recommended to answer them in order. Every file has a certain number of levels and each level can only be unlocked by completing the previous level. Each time you complete a level it will give you a score based on how many questions you've answered correctly. All levels will begin with a set of index cards that will serve as a refresher before the questions are given.

Level Mechanics

This section outlines the mechanics used in the game. This section shows the table of computation used in the technical part of the development.

- **Physics:** As the player talks to the NPC's their sprites will bounce up and down as well as show different emotions as the player goes thru their dialogue
- **Movement in the game:** The player navigates through the game by clicking buttons, icons, and menus. Movement consists of transitioning between different screens such as the Main Menu, Stage Selection Screen, Level Selection Screen, Accounting Workstation, Results Screen, and Customization Area.
- **Objects:** The player may tap or hold certain icons in order to pick them up
- **Actions and Interactions:** Tapping certain icons will allow the player to move from place to place as well as pick up objects, while tapping on any NPC will trigger a dialogue box to appear letting you interact with them.
- **Combat:** N/A
- **Economy:** Points earned through completing levels may be spent on cosmetics either for your character or their room. Certain cosmetics may only be unlocked on later levels/stages

- **Screen Flow:**



- **Game Options:** As the levels increase so does the difficulty. The Options Menu shall contain two settings namely Language and Sound settings.

- **Replaying and Saving:** Before and after each level will be saved. As well as when the player enters and exits the cosmetic shop .

- **Point systems:** After each level a rank will be given based on how many mistakes have been made and each rank will have a certain number of points and as stages increase so does the point multiplier

    S - 0 Mistakes - 100 Points
    A - 1-2 Mistakes - 80 Points
    B - 3-5 Mistakes - 60 Points
    C - 5-7 Mistakes - 40 Points
    D - 7 or more Mistakes - 20 Points

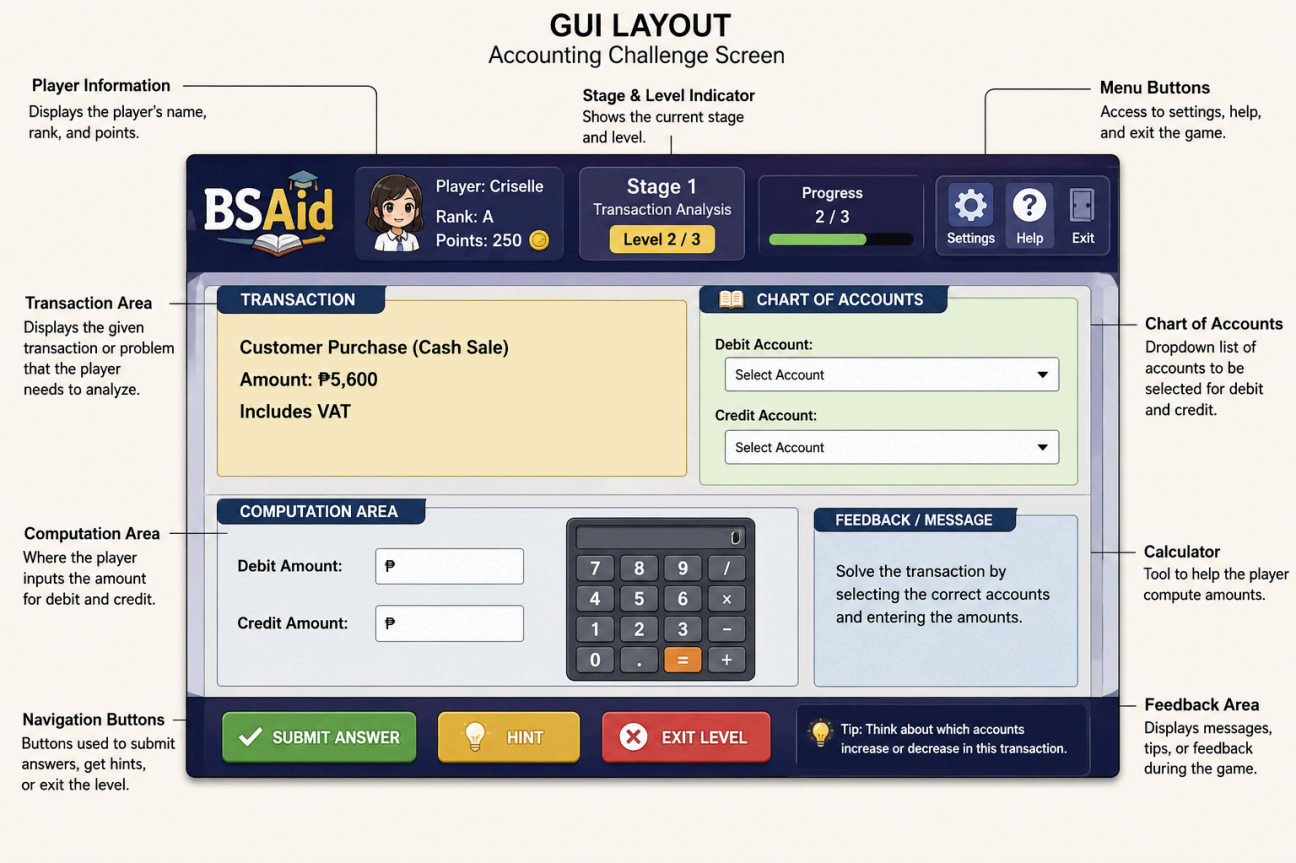
    Stage 1 - x1.0 Point Multiplier
    Stage 2 - x1.5 Point Multiplier
    Stage 3 - x2.0 Point Multiplier
    Stage 4 - x2.5 Point Multiplier

- **Cheats and Easter Eggs*:** N/A

User Interface

This section outlines the screen layout, visual elements, navigation controls, help features, and audio components used in BSAid.

- **GUI Layout:**



- **Camera Set Up:** Players do not need to move the camera during gameplay since it uses a fixed 2D screen view.
- **Camera Model:** Fixed screen layout, scrollable content pages, must also have a responsive design for different screen sizes
- **Icons:**
    - Home/Door - Main Menu/Exit Game
    - Settings - Game Options
    - Question Mark - Help
    - Calculator - Computation Tool
    - Coin - Points and Rewards
    - Green Button w/ Check - Submit Answer
    - Yellow Button w/ Light Bulb - Hint
    - Red Button w/ X mark - Exit Level
- **Game Control System:**

Action	Command
Select Option, Buttons, and Answers	Tap

Compute the Answer	Tap numbers & operations in the calculator
Classification of Accounts	Drag & Drop Accounts
Type Answer	On-screen Keyboard/Num Pad
Submit Answer	Tap Submit
Read explanation feedback	Scroll
Adjust Language & Sounds	Tap Settings

- **Help & Feedback System:**

  To support learning, the game provides:

  - Tutorial Mode
  - Hints & Guidance
  - Error Explanations
  - Correct Solution Review
  - Accounting Tips

  When players answer incorrectly, the system displays both the player's answer and the correct solution along with an explanation of the accounting principle involved.

- **Audio, Music, and Sound Effects:**

- https://youtu.be/PKdTziPZrhY?si=abIqTWMu0UVU3K_w - Achievement Sound
- https://youtu.be/DhHGDOgjie4?si=2N1XtbAs7oabhTb6 - Fail Quest/Explanation of Wrong Answer BGM [0:00 to 0:26]
- https://youtu.be/mjuymmDrz6o?si=xHcsDYhQTgRLcYwu - Main Menu & Gameplay Sound

Chapter 4 – EVALUATION AND TESTING

Hardware and Systems/Software Requirements

This section discusses the different software and hardware requirements to run

the game.

- **Development: (A)** Table specifications of hardware and systems requirements

  and (B) List of software used in the development.

- **User: (A)** Table specifications of hardware and systems requirements

Game Feature

This section discusses the different features of the game. Identify priority between integral and chrome features. List all the features and how they are used in the game. Provide screenshots and showcase all the key scenes of the end product it should include brief descriptions of the scene.

Game Screenshots

This section showcases the key scene of the end product it should include a brief description of the scene.

Testing

This section discusses the testing procedures of the game. Games are tested at different levels of development. It is the developers' duty to keep track of the functionality and integration of different components to assure the quality of the game project. This section lists preliminary testing of basic functions of the software and bug encountered after integration and being played by an audience with a fresh eye.

Functionality Testing: This subsection lists down all basic features that the

game project implements before reaching the beta stage of development.

Beta Testing: This subsection lists down all bugs/glitches encountered by the developers or reported by a testing audience and a discussion of fixes on these issues.

Multiplayer Testing*: If the game has a multiplayer feature, these features are tested and the results of testing are listed here. This includes but is not limited to establishing a connection, stress testing of servers, responsiveness, etc.

Summary of Evaluation Result

This section illustrates and discusses the result of the different testing methods used in evaluating the game. Provide appropriate tables, graphs, and other illustrations.